

Math 289, Fall 2026

Lecture 3

Warm-up. There is a light bulb inside a room and four switches outside. All switches are currently at off state and only one switch controls the light bulb. You may turn any number of switches on or off any number of times you want. How many times do you need to go into the room to figure out which switch controls the light bulb?

Problem 1. Define the sequence $(a_n)_{n \geq 0}$ by $a_0 = 0$, $a_1 = 1$, $a_2 = 2$, $a_3 = 6$, and

$$a_{n+4} = 2a_{n+3} + a_{n+2} - 2a_{n+1} - a_n, \quad n \geq 0.$$

Prove that n divides a_n for all $n \geq 1$.

Problem 2. Let $(x_n)_{n \geq 1}$ be a sequence of positive integers such that $\lim_{n \rightarrow \infty} x_n = \infty$. Does it follow that $\lim_{n \rightarrow \infty} x_n = \infty$?

Problem 3. Let $(a_n)_{n \geq 1}$ be a decreasing sequence of positive numbers converging to 0. Prove that the series

$$S = a_1 - a_2 + a_3 - a_4 + \cdots$$

is convergent.

Problem 4. Show that if the series $\sum a_n$ converges, where $(a_n)_n$ is a decreasing sequence, then

$$\lim_{n \rightarrow \infty} na_n = 0.$$

Problem 5. Let d be a real number. For each integer $m \geq 0$, define a sequence $\{a_m(j)\}$, $j = 0, 1, 2, \dots$ by the condition

$$\begin{aligned} a_m(0) &= \frac{d}{2^m}, \\ a_m(j+1) &= (a_m(j))^2 + 2a_m(j), \quad j \geq 0. \end{aligned}$$

Evaluate $\lim_{n \rightarrow \infty} a_n(n)$.